Detailed Certificate of Calibration

Equipment Calibrated		NIST Traceability Information
Description:	Six-Axis Force/Torque Sensor	NVLAP Lab Code: 200495-0
Manufacturer:	ATI Industrial Automation	Cal Standard: REF – 1000
Serial Number:	FT39887	Cal Standard: REF – 1001
Model:	Mini40 IP65	Cal Standard: REF – 1183
Calibration:	SI-20-1	Cal Standard: REF – 1119
Electronics:	DAQ	
Output Range:	±10V	
Gain Multiplier:	100%	

Equipment Condition and Notes: Factory new.

Calibration Results: Passed
Offset: Normal
Gain: Normal

Calibrated Ranges (±):

Fx	Fy	Fz	Tx	Ty	Tz
20 N	20 N	60 N	1 N-m	1 N-m	1 N-m

Measurement Uncertainty (95% confidence level, percent of full-scale load):

Fx	Fy	Fz	Tx	Ty	Tz
1.25%	1.25%	1.00%	1.25%	1.25%	1.50%

The above Measurement Uncertainty values are the maximum amount of error for each axis expressed as a percentage of its full-scale load.

Calibration Temperature: 22.2°±1.1° C (72°±2° F)
Temperature Compensation: none
Calibration Method: WI-FTP-026, DAQ Calibration Instructions

Date of Calibration: 06 Apr 2022

Certificate Date: 06 Apr 2022
Calibrated by: Shaquan Walden, Calibration Technician

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Quality Statement

As part of our commitment to quality, each ATI force/torque transducer undergoes rigorous accuracy testing. This process, which involves applying and verifying a rich set of loading cases designed to cover the transducer's entire six-axis calibrated range, is designed to ensure that your transducer meets the measurement uncertainties listed in this Certificate of Calibration.

Our transducers often exceed our rigorous quality standards for accuracy. To ensure that the published transducer measurement uncertainties listed on page 1 are met, uncertainty targets used during the calibration process are more conservative than the published values. Often, transducers perform exceptionally well in certain loading situations. This report summarizes the performance of your ATI F/T transducer in our factory tests. It can be thought of as a 'best-case scenario' snapshot of your transducer's performance under laboratory conditions, in a variety of loading situations. You can expect the accuracy of your transducer measurements to fall somewhere between its performance during testing and the measurement uncertainties listed on its calibration certificate.

Certificate of Calibration

This calibration is traceable to the National Institute of Standards and Technology (NIST).
ATI Industrial Automation (ATI) certifies that the listed product was calibrated in accordance with applicable ATI procedures. These procedures are compliant with the ISO 9001 standard to ensure that the listed product is within ATI's quality specifications. To meet this level of accuracy any loads must be correctly aligned to the transducer origin and the transducer must be robustly mounted to a surface of sufficient material strength and rigidity.

Note: If this is a recalibration of a legacy transducer that does not have precision locating features (such as dowel holes), users may experience additional error in measurements due to inexact mounting location. Precision locating features are highly recommended for best accuracy and can be added by ATI upon request.

The *Calibration Accuracy Section* contains several tables of data. The *Calibrated Ranges (±)* table lists the transducer's rated range for each axis. The *Applied Loads* table lists the loads applied during calibration and testing. The *Full-Scale Error* table shows the sensor system's measurement error as a percentage of full scale for each axis in each loading case. The *Offset Report* table, if included, shows transducer readings during offset adjustment and associated control limits. The *Gain-Check Report* table, if included, shows verification of the transducer's sensitivity and associated control limits. If included, the *Before and After Report* table shows a loading case relating the transducer's performance as received to its performance after recalibration.

For best accuracy, be sure to use your transducer's precision location features, and mount your transducer to a stiff surface. If an ongoing guarantee of sensor accuracy is important to you, we recommend that your sensor be tested annually. Contact your ATI Industrial Automation distributor to schedule recalibrations.

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Compliance Information - ISO 9001, A2LA, ANSI Z540, and ISO 17025

ATI Industrial Automation Inc. is an ISO 9001:2015 certified company. ATI's manufacturing facilities are not certified under A2LA, ANSI Z540, or ISO 17025, but our applicable processes are in compliance with these standards. As the designer and Original Equipment Manufacturer (OEM) of our Force/Torque Sensor products, we use validated, product-specific, proprietary manufacturing processes (including software), which no other facilities are capable of replicating. As the OEM of our Force/Torque Sensor products, ATI is not required to be A2LA, ANSI Z540, or ISO 17025 certified under the scope of our manufacturing operations. Parties interested in maintaining compliance with A2LA, ANSI Z540, or ISO 17025 standards as it pertains to their organization will find that our Force/Torque Sensor products meet several common exemption criteria specified in the aforementioned standards. In addition, 100% of our Force/Torque Sensor products are subjected to a testing and calibration sequence, specific to each sensor type, prior to shipping. When a part is returned as part of regularly scheduled maintenance (calibration) or a quality related-issue, it is re-processed through the same OEM testing and calibration sequence. All testing and calibration equipment used in our facilities are maintained in accordance with internal ISO 9001 compliant procedures and are traceable to NIST.

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Calibration Accuracy Section
Sensor System FT39887, Mini40 IP65/SI-20-1
Force units: N; Torque units: N-m

Calibrated Ranges (±)					
Fx	Fy	Fz	Tx	Ty	Tz
20	20	60	1	1	1

	Applied Loads					
	Fx	Fy	Fz	Tx	Ty	Tz
1	0.000	7.784	0.000	-0.790	0.000	0.000
2	-7.784	0.000	0.000	0.000	-0.790	0.000
3	0.000	-7.784	0.000	0.790	0.000	0.000
4	7.784	0.000	0.000	0.000	0.790	0.000
5	0.000	17.793	0.000	-0.541	0.000	0.000
6	-17.793	0.000	0.000	0.000	-0.541	0.000
7	0.000	-17.793	0.000	0.541	0.000	0.000
8	17.793	0.000	0.000	0.000	0.541	0.000
9	0.000	6.116	0.000	-0.047	0.000	-0.700
10	0.000	6.116	0.000	-0.047	0.000	0.700
11	-6.116	0.000	0.000	0.000	-0.047	-0.699
12	-6.116	0.000	0.000	0.000	-0.047	0.699
13	0.000	-6.116	0.000	0.047	0.000	-0.700
14	0.000	-6.116	0.000	0.047	0.000	0.700
15	6.116	0.000	0.000	0.000	0.047	-0.699
16	6.116	0.000	0.000	0.000	0.047	0.699
17	0.000	0.000	11.677	-0.892	0.000	0.000
18	0.000	0.000	11.677	0.000	-0.892	0.000
19	0.000	0.000	11.677	0.888	0.000	0.000
20	0.000	0.000	11.677	0.000	0.891	0.000
21	0.000	0.000	53.379	0.000	0.000	0.000
22	0.000	0.000	-53.379	0.000	0.000	0.000
23	0.000	0.000	-11.677	0.892	0.000	0.000
24	0.000	0.000	-11.677	0.000	0.892	0.000
25	0.000	0.000	-11.677	-0.888	0.000	0.000
26	0.000	0.000	-11.677	0.000	-0.891	0.000

Refer to page 2 for important information on regarding this report.

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	Full-Scale Error					
	Fx	Fy	Fz	Tx	Ty	Tz
1	0.06%	-0.30%	0.12%	0.01%	-0.17%	-0.33%
2	-0.29%	-0.05%	0.04%	-0.17%	0.13%	0.05%
3	0.05%	-0.42%	0.15%	-0.03%	-0.18%	-0.04%
4	0.09%	0.10%	0.10%	0.00%	-0.03%	0.26%
5	0.07%	-0.25%	0.15%	0.13%	-0.13%	-0.19%
6	-0.16%	-0.06%	0.09%	-0.02%	0.16%	0.20%
7	0.09%	-0.43%	0.13%	0.12%	-0.11%	-0.15%
8	-0.09%	-0.06%	0.08%	-0.05%	0.03%	0.20%
9	-0.04%	-0.15%	0.01%	0.15%	-0.05%	0.28%
10	0.11%	-0.20%	-0.01%	0.05%	0.11%	0.12%
11	0.05%	-0.07%	0.01%	0.12%	0.12%	-0.05%
12	0.13%	0.15%	0.00%	-0.01%	-0.44%	0.23%
13	0.10%	0.22%	0.00%	0.03%	-0.04%	-0.01%
14	-0.08%	0.16%	0.01%	0.25%	0.03%	-0.07%
15	-0.24%	0.02%	-0.01%	-0.02%	-0.02%	-0.01%
16	-0.30%	-0.11%	-0.01%	-0.02%	0.30%	-0.07%
17	-0.02%	-0.25%	0.06%	0.14%	-0.09%	0.06%
18	-0.15%	-0.03%	0.02%	0.20%	-0.13%	-0.09%
19	0.03%	-0.24%	0.00%	0.28%	-0.22%	-0.04%
20	-0.24%	-0.09%	-0.01%	0.31%	-0.12%	0.01%
21	0.08%	-0.56%	0.03%	0.15%	0.18%	-0.07%
22	0.05%	-0.59%	-0.05%	0.41%	-0.11%	-0.13%
23	0.00%	-0.07%	0.17%	-0.24%	0.29%	-0.01%
24	-0.28%	-0.11%	0.07%	-0.06%	0.25%	-0.10%
25	0.05%	-0.25%	0.13%	-0.14%	0.15%	0.14%
26	-0.02%	-0.05%	0.09%	0.17%	0.08%	0.17%

Refer to page 2 for important information on regarding this report.

	Offset Report					
	Fx	Fy	Fz	Tx	Ty	Tz
F/T Offset	-0.0369	-0.0181	0.1477	-0.0004	0.0042	-0.0016
	SG0	SG1	SG2	SG3	SG4	SG5
SG Offset	0.0202	0.0144	0.0233	0.0046	0.0270	0.0172
±SG Limit	0.4883	0.4883	0.4883	0.4883	0.4883	0.4883

Offsets are measured in a unique configuration not available to the user.
Refer to page 2 for important information on regarding this report.

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	Gain-Check Report					
	SG0	**SG1**	**SG2**	**SG3**	**SG4**	**SG5**
Lower Limit	*0.8000*	*0.8000*	*0.8000*	*0.8000*	*0.8000*	*0.8000*
Lower Output	0.8494	0.8200	0.9007	0.8873	0.9454	0.8058
Upper Output	0.8543	0.8421	0.9036	0.9191	0.9775	0.8352
Upper Limit	*1.0000*	*1.0000*	*1.0000*	*1.0000*	*1.0000*	*1.0000*

Gain readings are measured in a unique loading configuration.

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